Solar cells, batteries and the grid: from buildings to whole countries

Technologies in use and not yet viable · Storage by building type · Equipment ecosystem · Demand 2005–2045 · Policy debates · Transmission models from 8 countries · End-of-life

Audience
Policy planners / municipalities · engineers

Currency
USD and THB (assumed 33 THB/USD)

Symbol
◇ = baseline data / estimate not yet checked against the original source

Six key findings

01 · CELL TECH

TOPCon ~67–87%

of 2025 cell production (ITRPV lowest, CPIA/Exawatt highest). PERC is disappearing, and perovskite tandem stays commercially limited before 2030.

02 · STORAGE COST

$70/kWh ≈ 2,310 ฿

Stationary battery pack price in 2025, down 45% year on year (BNEF). Turnkey systems average $117/kWh, and sodium-ion begins commercial deliveries in 2026.

03 · DEMAND

+3.6%/yr

The IEA expects world electricity demand to grow 3.6% a year over 2026–30 (2.8% in the previous decade). Data centres rise from 485 to 950 TWh by 2030.

04 · BOTTLENECK

Transmission and flexibility

Chile curtailed ~18% of renewables (2024) · Rajasthan curtailment peaked at 51.5% · Germany set a record for negative-price hours (2025).

05 · THAILAND

9.9 → 39 GWp

Solar capacity in 2024 towards the AEDP 2037 target. Ember estimates that another 32 GW of solar plus 6 GW/15 GWh of BESS would be cheaper than the current plan.

06 · END-OF-LIFE

60–78 Mt

Cumulative global panel waste by 2050 (IRENA). Thailand has no EPR law specific to panels ◇, so the system should be designed now.

01

Technology

Cell types in use and not yet viable · battery chemistries · matching to 6 building types · equipment ecosystem

Cell types actually on the market (2025–26)

01 · IN USE

Technology	2025 share	Module efficiency	Strengths / limits	Meaning for Thailand
TOPCon n-type	**67–87%** ITRPV ↔ CPIA/Exawatt	up to ~24.1%	The industry's mainstream standard and will lead the market through the decade, though its share will slowly fall	The default for solar farms and factory rooftops
Back-contact BC / XBC	**~5–6%** expected ~33–40% in 2030–35	~25%	No front busbars shading the cell, so power per area is high, but the price is higher	Suits space-limited roofs such as offices and urban hospitals
HJT heterojunction	**~5–8%**	~23.8%	Heat tolerant with slow degradation (~0.25–0.30%/yr ◇) but uses a lot of silver and is a niche market	An advantage in a hot climate, and a base for future tandems
PERC p-type	**~14–18%** falling fast	20–21.5%	Lowest cost per unit, but production lines are being retired	Watch for old stock dumped on the home market and warranties from makers that may exit
Thin-film CdTe ◇	Minority	~19–20% ◇	Handles heat and partial shade well, but contains cadmium and needs a recycling system	Usable at utility scale if tied to panel-waste requirements

Sources: TaiyangNews/ITRPV·CPIA·Exawatt 2026 · TaiyangNews 2025 · Accelerate Solar 2026 (module efficiency from listings)

Technologies "not yet viable" for wide investment

PEROVSKITE–SILICON TANDEM

35.5%
Lab cell record (LONGi, Jan 2026, certified by ESTI); a small cell

33%
260.9 cm² cell (NREL, Jun 2025)

24–29%
Commercial module efficiency today; GCL has a 500 MW line

Why not yet viable: moisture and heat stability not proven over 25 years · contains lead · measurement and insurance are not yet standardised · every projection agrees it stays limited before 2030.

For Thailand: run pilots on government or university buildings that collect hot, humid climate data, then consider wide procurement after 3–5 years of field data.

Flexible / lightweight tandem
A 2026 Nature paper reports a 33.6% cell that keeps 91% after 5,000 bends, but it is still at research stage. Suits old factory roofs that cannot take weight

Single-junction perovskite
~27% in the lab, but service life is still the main question

BIPV / solar façade ◇
Cost per kWh is much higher than rooftop; walls in the tropics get little light. Only viable when it replaces cladding you would buy anyway

CPV / III-V ◇
Needs strong direct sunlight, while Thailand has lots of cloud and humidity. Mostly used in space today

Organic PV ◇
Efficiency and lifetime still low; only for IoT or indoor devices

Sources: pv magazine, Jan 2026 · TaiyangNews 2026 · GreenFuel Journal 2026 (secondary)

Battery chemistries for solar systems

Stationary pack 2025
$70/kWh
≈ 2,310 ฿ · −45% YoY

Average turnkey BESS
$117/kWh
≈ 3,861 ฿ · −31% YoY

2 h / 4 h system
$124 / $110
≈ 4,092 / 3,630 ฿

LFP vs NMC pack
$81 / $128
Average across all uses

Chemistry	Characteristics	Where used	Status
LFP	Safer than NMC with no cobalt or nickel; cheapest; China makes almost all of world demand	Homes, C&I, utility, data-centre UPS	Mainstream
Sodium-ion	CATL cell ~160 Wh/kg, >15,000 cycles, works from −40 to 70 °C; same size as the 587 Ah LFP cell, so it is easy to swap	Grid storage with a 25–30 year life, data centres	Deliveries from Sep 2026
NMC	High energy density, but costlier with more thermal-runaway risk	EVs, space-constrained sites	Shrinking role in stationary
Flow (VRFB) ◇	Energy scales separately from power; long cycle life, but high CAPEX and a large footprint	Long duration 6–10 h at substation level	Not yet viable versus 4 h LFP
Lead-acid ◇	Low upfront price, but short cycle life and low DoD make lifetime cost high	Legacy UPS, rural off-grid	Being replaced by LFP

Sources: BNEF Battery Price Survey, Dec 2025 · BNEF ESS Cost Survey 2025 · ESS News, Apr 2026 · CarNewsChina, Jun 2026

PV + storage by building type

01 · INTERACTIVE — FACTORY SELECTED

Home | Factory drill-down | Hospital drill-down | School drill-down | Office | Data centre

LOAD PROFILE

High, fairly steady daytime load over 1–3 shifts, with demand charges and RE100 needs for export customers and CBAM

TYPICAL PV

0.5–10 MWp on roofs, car-park roofs and floating on ponds, with string inverters, connected at 22/33 kV

SUITABLE STORAGE

C&I BESS of 1–2 h for peak shaving to cut demand charges and support machine ramp, and as UPS for processes that can't stop

Key equipment

PCS, an EMS linked to the plant's SCADA, power-quality meters, BESS cabinet fire suppression and protection relays

Mechanisms in Thailand

Direct PPA / UGT (pilot 2,000 MW), zero-capex ESCOs and BOI incentives. FiT round 2 also has a PV+BESS quota (1,580 MW + 600 MW)

Watch-outs / debate

Old roofs may not take the weight, so check the structure first. 15–25 year PPAs may outlast land leases, and wheeling charges of about 0.3–0.6 ฿/kWh ◇ affect viability

System sizes are indicative ranges from the author's analysis ◇ · Thai mechanisms: Green Energy Thailand 2026 · Ember 2025 · UGT summary · the original report is interactive; the PDF shows the Factory view

The equipment ecosystem you need

Generation

Modules + mounting / tracker
Single-axis trackers for solar farms, floating systems for dams

String inverter
For C&I roofs and mid-size farms

Central inverter
For farms of 10 MW and up

Optimiser / micro-inverter
For shaded roofs; helps with rapid shutdown

Storage

Hybrid inverter / PCS
Bidirectional conversion and backup support

BMS
Monitors and balances cell SOC/SOH

Thermal + fire safety
Liquid cooling, gas detection and deflagration vents, vital in hot climates

Grid-forming inverter
Sets its own voltage and frequency; for weak grids, microgrids and islands

Grid connection

Smart meter / AMI
A precondition for TOU, net billing and German-style compensation

Protection relay + anti-islanding
Stops backfeed while lines are being repaired

Transformer / switchgear
Must handle reverse power flow

ATS + genset
For hospitals and data centres

Control / data

EMS / BEMS
Peak shaving and load shifting at building level

DERMS + SCADA
Lets the utility see and dispatch DER

VPP aggregator
Pools home batteries to sell services to the grid

Forecasting
Forecasts sunlight and load at feeder level

Proposal for Thailand: set communication standards (such as IEEE 2030.5 / IEC 61850 ◇) and make a smart meter a condition of selling power back, so VPP and DERMS can really happen later.

02

Systems and demand

A 6-layer architecture from buildings to the region · statistics 2005–2025 · scenarios to 2045 · expansion to secondary cities

How it connects: building → city → country → region

Layer	Level	Assets / voltage	Storage role and control	Responsible (Thailand)
L5	**Cross-border**	500 kV and HVDC interconnectors · ASEAN Power Grid ◇	Trade Lao hydropower against daytime solar and share reserves	EGAT, ERC, government, neighbouring countries
L4	**Country**	500/230 kV backbone · national control centre	Dispatch, reserves, market/price; large BESS for frequency response	EGAT (system operator), ERC, NEPC
L3	**Region**	115/230 kV · solar farms, floating PV on dams	BESS at substations (India-style pooling stations) to cut curtailment	EGAT, SPP/IPP
L2	**City / province**	22/33 kV and 115 kV distribution · feeders · transformers	Hosting capacity, DERMS, feeder BESS and community solar ← **the biggest gap**	PEA, MEA, municipalities
L1	**Microgrid / estate**	Industrial estates, campuses, hospitals	Islanding, peak shaving and Direct PPA/UGT	Estate developers, ESCOs, universities
L0	**Building / prosumer**	Rooftop PV, hybrid inverter, home battery, EV	Maximise self-use; join VPP and V2G in future	Households, installers

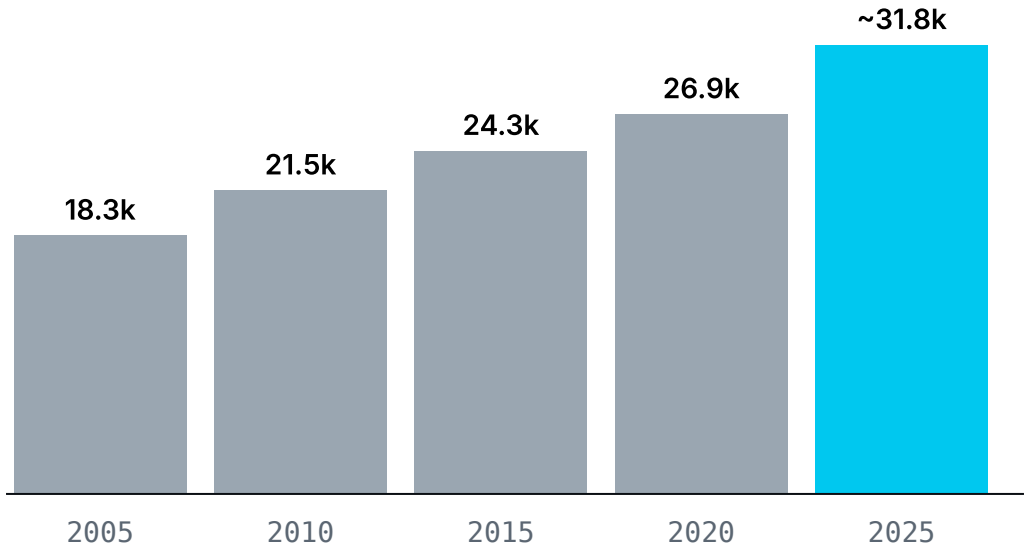
Flowing up: surplus daytime energy, meter data and flexibility services

Flowing down: price signals, curtailment orders and night-time power

Thailand 2024: PV 9,942 MWp, split into ground-mounted 6,334, rooftop 3,310 and floating 281 (IEA-PVPS)

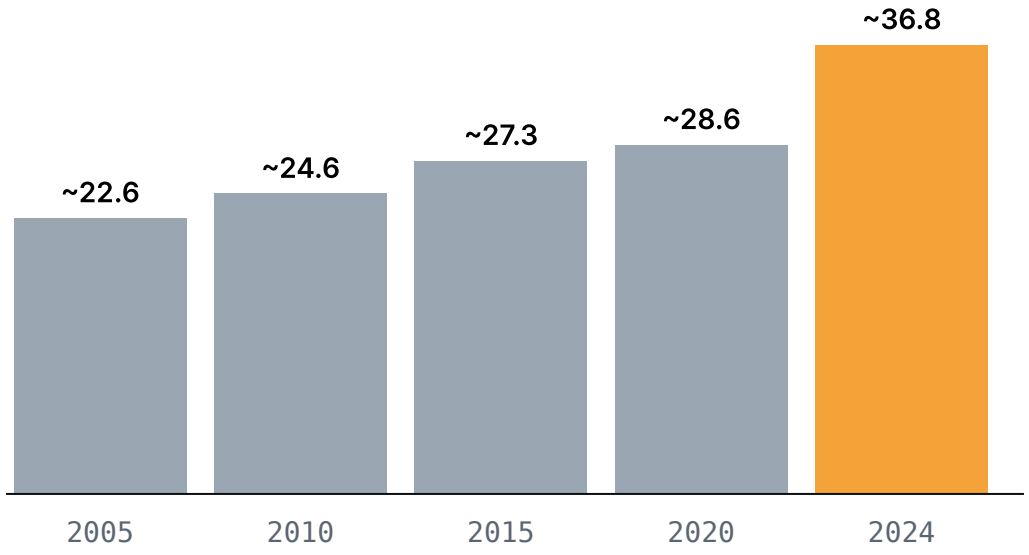
Electricity demand over the past 20 years (2005–2025)

WORLD · TWh/yr (rounded ◇)



Average growth of 2.8%/yr over the past decade · 2024 was +4.0% on heatwaves · 2025 +3% · solar was 8.7% of generation in 2025 (6.9% in 2024) · world solar capacity went from 1 TW to 2 TW in two years (2022→2024)

THAILAND · PEAK DEMAND GW (approx. ◇)



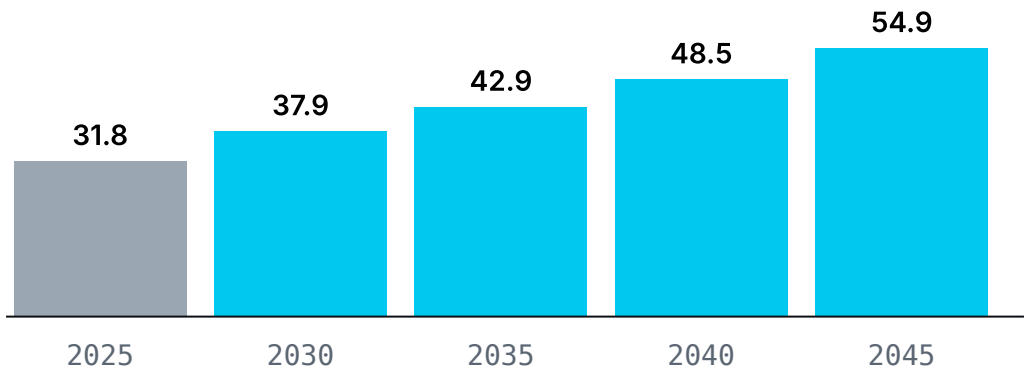
Total electricity about 128 → 205 TWh ◇; the peak grew faster than energy because of air conditioning and heat. Note that this series is from the author's recollection and should be checked against EPPO/EGAT statistics before use.

Sources: IEA Electricity 2026 · Ember GER 2025 · Ember World 2025 · Thailand: ◇ estimates to be confirmed with EPPO

The next 20 years: scenarios to 2045

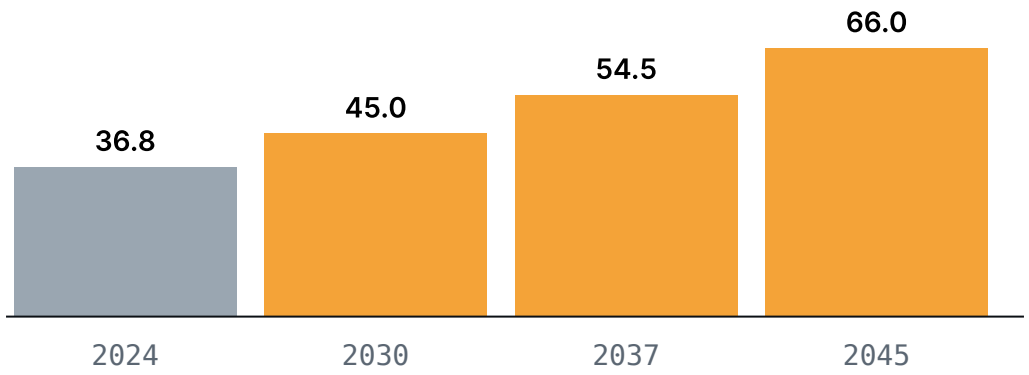
Base case follows trend: IEA 3.6%/yr, then easing to 2.5% · Thailand follows the RPDP base

WORLD · THOUSAND TWh/yr



Assumption: IEA +3.6%/yr over 2026–30 (≈ +1,100 TWh/yr); beyond that, the author's extrapolation ◇

THAILAND · PEAK DEMAND GW



Assumption: RPDP 2024 base ~54.5 GW (2037) and a data-centre case of 57.9 GW (Ember); 2045 is an extrapolation ◇

Data centres: 485 → 950 TWh by 2030 (~3% of the world); the queue to connect large loads grew 3.5 times over 2024–26

Buildings: 49% of growth over 2025–30, nearly half of it from air conditioning, data centres and heat pumps

Thailand: Direct PPA pilot for data centres of 50 MW and up, 2,000 MW in total · UGT 2,000 MW

Sources: IEA Electricity 2026 · IEA Energy & AI via Enlit · Ember Thailand 2025

Preparing for growth from major to secondary cities

Drivers

- The EEC and new estates (Chonburi, Rayong, Chachoengsao) attract data centres and factories that want RE100 power
- Motorways and double-track rail make secondary cities reachable, such as Korat, Khon Kaen, Udon Thani, Hat Yai and Chiang Mai
- Cheaper land suits solar farms and BESS at substations
- Community Solar (launched 27 Oct 2025) targets 1,500 MW, capped at 10 MW per site, buying at 2.25 ฿/kWh

Constraints

- Hosting capacity of 22 kV feeders in the provinces is limited; midday backflow pushes voltage too high
- Expanding PEA transformers and substations takes years
- Municipalities lack investment budget, load data and engineers
- Too few certified installers and inspections raise the fire risk from DC wiring and batteries

Planning principles

1. Publish a feeder-level **hosting capacity map** (as used in California and Australia ◇)
2. Use **grid-ready** city plans that reserve land for substations and BESS before the city grows
3. Require new public buildings to be **PV-ready** in both roof structure and conduits
4. Make provincial hospitals the **microgrid** core of city resilience
5. Use flexible connections: connect faster in exchange for accepting curtailment

Community Solar: Green Energy Thailand 2026 (citing Hunton Andrews Kurth) · the rest is the author's analysis ◇

03

Debates and stakeholders

7 debates where planners must pick a side · a value-chain map of who gains and who is affected

Debates in design, planning and investment

Issue	For	Against	Proposal ◇
Rooftop vs farm	Close to load, no extra transmission, spreads benefits to people	Higher cost per watt; creates backflow in distribution that the utility can't see	Do both, tying rooftop to smart meters and DERMS
Batteries now vs wait	Germany no longer pays during negative prices, so systems without batteries lose income	Pack prices fell 45% in one year, so waiting has value, and sodium-ion is coming	C&I with demand charges can install now; homes should wait for TOU signals
Grid first vs generation first	India and Chile show that without lines, curtailment is huge	Lines take 5–10 years and risk being built and left unused	Plan RE zones with transmission, and use BESS as a bridge while waiting
Net metering vs self-use	Incentives attract installs; Vietnam raised the export cap from 20% to 50%	Utility revenue falls and costs shift to people without solar (death spiral)	Net billing by TOU with a clearly separate grid charge
New gas vs PV+BESS	Ember estimates Thailand needs no 2 GW of new gas and saves $1.8B by 2037	The rainy season has little sun; firm capacity and security are needed	Run an hourly reliability study and publish the data
Cheap imports vs local production	China's overcapacity makes panels and batteries the cheapest in history	Supply-chain risk, trade measures and inverter cyber security	Produce locally only in high-value steps such as integration and EMS
Farmland vs floating	Floating PV on dams (EGAT 45 MW at Sirindhorn, 24 MW at Ubol Ratana) needs no land	Higher cost and open questions about aquatic ecosystems	Use agrivoltaics and dams that already have transmission

Sources: Ember 2025 · Viet Nam News 2026 · IEA-PVPS Thailand · Earth Energy Log (secondary)

Value-chain map: who gains, who is affected

	Upstream Panel, battery and inverter makers/importers	**Developers / installers** EPC, technicians, ESCO	**Finance** Banks, BOI, ENCON Fund	**Grid** EGAT, PEA, MEA, ERC	**Users** Homes, factories, hospitals, schools, data centres	**Downstream** Recyclers, Dept. of Industrial Works, communities
+ GAIN	Growing BESS and inverter market; chance for local assembly	Installation and O&M work spreads to every province, including VPP	Assets with clear cash flows, such as 20–25 yr PPAs and green bonds	Less investment in peaker plants and flexibility services from DER	Lower bills, RE100 for exporters and hospitals with disaster backup	Recovered materials (silver, copper, lithium) and a new industry
– AFFECTED	Undercut by imports; standards change fast	Small, uncertified firms risk fires and can't honour warranties	Curtailment risk and hard-to-predict merchant prices	Falling revenue (death spiral) and the cost of distribution upgrades	People without roofs (renters, condos) may end up paying grid costs instead	Hazardous waste reaches scrap shops, and informal workers are exposed to toxins

Grouping and impacts are the author's analysis ◇ and should be checked through real stakeholder consultation

04

Models from 8 countries

Germany · Australia · California · China · India · Japan · Vietnam · Chile

Transmission and policy models

04 · INTERACTIVE — GERMANY SELECTED

Germany | Australia | California | China | India | Japan ◇ | Vietnam | Chile

GERMANY · Solarspitzengesetz

Price signals + smart meters for rooftop

~117 GW
Cumulative PV installed by end-2025

573 / 1,100+
Negative-price hours in 2025; the two sources disagree (457 h in 2024)

60%
Export cap for new systems without a smart meter

Structure / mechanism

Solar Peak law in force Feb 2025: new PV systems over 2 kW receive no EEG payment in 15-minute slots with negative prices, compensated by extending the end of the support term

Problem faced

Midday sun floods the market, negative-price hours rise every year, and the value of solar power (capture price) falls

Lesson for Thailand

Thailand should drop flat all-day FiT, use TOU-based net billing and make smart meters a condition, which makes home batteries pay for themselves

Sources: pv magazine · Copernica · 3E

The original report is interactive; the PDF shows the Germany view

Models Thailand should adopt

1 **Transmission first + BESS at substations**

From **India / Chile / China**: zone RE with a transmission plan and install 2 h BESS at substations while waiting

2 **Price signals instead of flat FiT**

From **Germany**: use TOU and time-reflective net billing, pay nothing when the system is oversupplied, and make smart meters a condition

3 **Subsidise batteries paired with solar**

From **Australia**: a declining per-kWh rebate, with home batteries required to join VPPs

4 **Capacity contracts for storage**

From **California**: pay resource adequacy so BESS covers the evening peak, and enlarge Thailand's PV+BESS FiT

5 **Direct PPA with clear wheeling charges**

From **Vietnam**: open DPPA to estates, data centres and EV charging, and gradually raise the surplus export cap

6 **Avoid a boom with no grid behind it**

From **Vietnam 2019–20 / Japan** ◇: set FiT quotas from each region's real hosting capacity

05

End-of-life

Recycling technology and economics · EPR laws abroad · the situation in Thailand

Waste volumes, technology and recycling economics

60–78 Mt
Cumulative global panel waste by 2050 (IRENA)

>$15B
Value of recoverable materials by 2050 (IRENA 2016); some sources cite $8.8B

~30 yrs
Panel life used in projections; early replacement would bring waste forward

~0.6–2.5 Mt
Thailand: future waste from 10 GWp to 39 GWp (at ~60–65 t/MW) ◇

Solar panels

Mechanical: strip the aluminium frame and crush the glass; low cost but low-value materials ◇

Thermal / chemical: separate the EVA and recover silver, silicon and copper; higher value but needs heavy investment ◇; research uses organic acids and ultrasound (Springer 2026)

Economics: treatment costs in Europe are about €0.10–0.50/kg, while panels using less silver lower the recoverable value too

Lithium batteries

Pyrometallurgy: smelting recovers Co/Ni/Cu but loses some lithium ◇

Hydrometallurgy: acid leaching recovers lithium well and is the mainstream in China ◇

Direct recycling: restores cathode material directly; still at pilot stage ◇

The LFP problem: no Co/Ni, so recovered value is low; it needs a mandate or an upfront fee

Sources: IRENA/IEA-PVPS 2016 · Springer 2026 · SurgePV 2026 (secondary)

EPR laws abroad and the situation in Thailand

Where	Rule
EU · panels	The WEEE Directive was the first to apply specifically to panels: every producer selling in the EU must fund collection and recycling, with targets of 85% collection / 80% recycling
EU · batteries	Battery Regulation 2023/1542 sets minimum recycled content in new batteries and lithium recovery targets, with a battery passport ◇
India ◇	E-Waste Rules 2022 cover solar panels, and there are EPR rules for batteries
Japan / US	General laws with hazardous-substance testing and disposal guidelines; some US states have stewardship laws ◇
China ◇	EV battery tracking and a list of certified recyclers; a framework for panels is being developed

THAILAND · TO BE CHECKED ◇

No EPR law specific to panels or batteries has been found. Waste is handled as industrial waste, and the draft WEEE law on electrical products was not yet in force at the time of writing.

Proposals

1. Charge an **upfront fee** at import, per kW and per kWh
2. Run a **panel and battery registry** tied to PEA/MEA connection permits
3. Set up **regional collection centres** near large solar farms
4. Write decommissioning terms into **PPA contracts** from day one

Sources: IRENA via AA · SurgePV 2026 · IRENA/IEA-PVPS PDF · rows marked ◇ = baseline data not yet re-checked

Sources

ACCESSED OCT 2026

IEA — Electricity 2026

Ember — Global Electricity Review 2025

BNEF — Battery Price Survey 2025

pv magazine — LONGi 35.5%

Copernica — DE negative prices

Modo Energy — CAISO 2026

OIES — China power sector 2026

Viet Nam News — Decree 243

IEA — Key Questions on Energy and AI

Ember — Thailand cost-optimal pathway

Energy-Storage.News — BNEF ESS cost

ESS News — CATL sodium-ion

3E — Netzpaket / Solarspitzen

SEIA — H1 2026 records

Mercom — India Q1 2026

Wikipedia — Curtailment in Chile (secondary)

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TaiyangNews — cell tech shares

IEA-PVPS — Thailand 2025

pv magazine — AU CEC report H1 2026

pv magazine — China 315 GW

Business India — Rajasthan

IRENA — End-of-life PV panels

Limitations: Japan data, Thailand's historical figures and the status of Thai EPR law were not checked against the original sources in this round (◇) · the 2045 scenario is the author's extrapolation · exchange rate assumed at 33 THB/USD · English edition of the Thai report; links are in the Thai PDF